

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12.				From % composition ratio C: H : O $\frac{61.2}{12} : \frac{6.1}{1.01} : \frac{32.7}{16} = 5.1 : 6.04 : 2.04$ (1) 2.5 : 3 : 1 → 5 : 6 : 2 Empirical formula is C₅H₆O₂ (1) From mass spectrum <i>M_r</i> is 98 (1) Molecular formula is C₅H₆O₂ (1) Identification of one fragment from m/z value (1) From reaction with sodium carbonate X is a carboxylic acid/ contains CO₂H (1) From ¹³C spectrum There are 5 different carbon environments (1) From reaction with bromine 320 g of bromine is 2 mol (1) X contains 2 (C to C) double bonds (1) X is CH₂=CHCH=CHCO₂H (accept any isomer with 2 double bonds, 5 carbon environments and CO₂H) (1)	1	2			2	
				Question 12 total	1	8	1	10	2	0